Summary in Graph

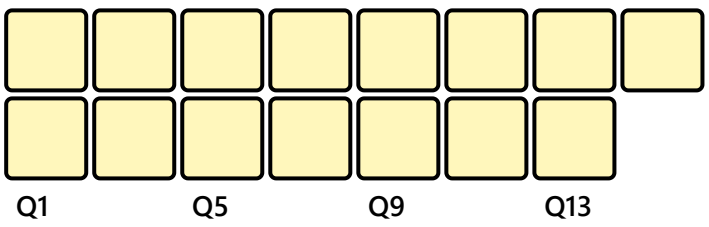
Exam Summary (GO Classes GATE CS | C-Programming | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (60) out of 604 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Programming in C

As shown below, the file $f.c$ defines a function f whose code refers to the variable x that is not a local variable or a parameter. The file $g.c$ defines a function g whose code refers to the variable x that is not a local variable or a parameter.

```
// file f.c
// declaration for x

void f(int a){
    ...
    x = ...
}
```

```
// file g.c
// declaration for x

void g(char* p){
    ...
    x = ...
}
```

The following table below shows how the variable x is defined in each file. For each row in the table, indicate in Column (c) whether the two functions at run-time refer to the SAME identifier (memory location) or to DIFFERENT identifiers (memory locations)

Column A declaration in f.c	Column B declaration in g.c	Column C SAME or DIFFERENT
int x;	extern int x;	**SAME**
int x = 0;	static int x;	
static int x;	static int x;	
static int x;	extern int x = 0;	

The first row of column C has been filled for hints. What will be in the 2nd, 3rd, and 4th row of Column C, respectively?

A. SAME, DIFFERENT, DIFFERENT
B. SAME, SAME, SAME
C. DIFFERENT, DIFFERENT, DIFFERENT
D. DIFFERENT, SAME, DIFFERENT

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the below program written into 2 files.

f1.c

```
#include <stdio.h>
extern int x;

void fun(){
    printf("%d", x);
}
```

f2.c

```
#include <stdio.h>

static int x;
void fun();

main() {
    fun();
    printf("%d", x);

}
```

Note that compilation of f1.c and f2.c happens independently of each other.

Which of the following is TRUE?

A. f1.c and f2.c can be compiled independently.

B. f1.c will produce a compilation error since variable x does not get any memory, and we are using it in printf.
C. f1.c and f2.c can be compiled and linked. The output of the program will be 00.
D. f2.c will produce a compilation error since function fun() is not defined in f2.c.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Consider a function is_greater() given below. Mark all option(s) which return 1.

```
int is_greater(unsigned int x, unsigned int y)
{
    if(x-y > 0)
        return 1;
    else
        return 0;
}
```

A. is_greater$(1, 2)$
B. is_greater$(-1, -2)$
C. is_greater$(-2, -1)$
D. is_greater$(-1, 0)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Which of the following(s) will evaluate to true

A. $0 \;\&\&\; 0 == 0$
B. $0 \;\&\&\; 1 == 0$
C. $1 \;||\; 0 == 0$
D. $1 \;||\; 1 == 0$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output printed by the following program?

```c
#include <stdio.h>
int main()
{
    int i = 1;
    for(i = 0; i<10; i+3)
    switch(i)
    {
        case 3:
            printf("Hi. This is case 3");
            break;
        case 6:
            printf("Hi. This is case 6");
            break;
            break;
        default:
            printf("Hi. This is default");
            break;
    }
    return 0;
}
```

A. Hi. This is case 3
B. Hi. This is case 6
C. Infinite Execution
D. Hi. This is default

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the value printed by the following program?

```c
int i = 10;
i = !i > 14;
printf("%d", i);
```

A. 10
B. 14
C. 0
D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output?

```c
int a = 4, b =15, c =29;
if(c>b>a)
    printf("TRUE");
else
    printf("FALSE");
```

A. TRUE
B. FALSE
C. Syntax Error
D. Compilation Error

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

If x, y, z, and w are declared as integer variables, which of the following expressions are NOT valid in C?

A. $x + 5 = y + z$
B. $x = y = z + w$
C. $x = (z - y) == w$
D. $w = x! = y \;\&\&\; z$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following C program?.
Here ? : is a ternary operator which is right to left-associative.

```c
#include <stdio.h>
int main() {
    int i =2, j =1;
    if (--j ? ++j? i : i++ : i--)
    printf("%d%d\n", i, j);
}
```

A. 1 0
B. 2 1
C. 1 1
D. 2 0

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What is the value printed by the following program?

```c
int main()
{
    int a = 0, b = 1, c = 0, d =-1;
    d = (++a) && (--b) && (++c) ||(--a);
    printf("%d", d);
}
```

A. 0
B. 1
C. -1
D. 2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

The statement

```c
while ( --counter >= 1 )
counter % 2 ? printf("A") : printf("B");
```

can NOT be rewritten as

A.

```
while ( --counter >= 1 )
    if ( counter % 2 )
        printf("A");
    else
        printf("B");
```

B.

```
while ( counter >= 1 )
    if (counter % 2)
        printf("A");
    else
        printf("B");
--counter;
```

C.

```
while ( counter > 1 )
{
    --counter;
    if ( counter % 2 )
        printf("A");
    else
        printf("B");
}
```

D.

```
do
{
    --counter;
    printf( counter % 2 ? "A" : "B" );
} while ( counter >= 2 );
```

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12

Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Professor Greg wants to give assignments to his students to write a C program for the $\mathrm{XOR5}$ function. $\mathrm{XOR5}$ is a function that returns true if and only if $\mathrm{EXACTLY}$ one of the integers A and B is 5. Using which of the following condition(s), $\mathrm{XOR5}$ can be implemented?

A. $(A == 5)! = (B == 5)$
B. $!(A == 5)! =!(B == 5)$
C. $(A == 5)?!(B == 5) : (B == 5)$
D. $((A == 5)||(B == 5))\ \&\&\ !((A == 5)\ \&\&\ (B == 5))$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13

Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Let the variables x, y, and z, be of type int and assigned some values. The variable $b1$ is initialized as follows:

```
int b1 = (x > z) && (y >= z);
```

Variables b2, b3, b4, and b5 are initialized below.

```
    int b2 = (x <= z) || (y < z)
    int b3 = !((x < z) || (y <= z))
    int b4 = !((x < z) && (y >= z))
    int b5 = ((x > z) && (y > z)) || ((x > z) && (y == z))
5.
```

Variables $b2, b3, b4$, and $b5$ ALWAYS, SOMETIMES, or NEVER takes the same value as $b1$.
Which of the variable(s) SOMETIMES takes the same value as $b1$?

A. $b2$
B. $b3$
C. $b4$
D. $b5$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following two $.c$ files which both include the same $.h$ file:

```
// a.h
(1)         int inc(int x) { return x + 1; }
(2)         int x;
(3)         int y;
(4)         int z;
(5)         void b(void);
```

```
// a.c
#include "a.h"
#include <stdio.h>

(6)         int y = 1;
(7)         int w;
(8)         int v = 5;

int main()
{
    x = inc(0);
    z += 4;
    b();
    printf("x = %d y = %d z = %d "
           "w = %d v = %d\n",
            x, y, z, w, v);
    return 0;
}
```

```
// b.c
#include "a.h"

(9)         int w = 4;
(10)        int v = 5;

void b() {
    x = inc(x);
    y++;
    z--;
    v++;
}
```

When compiled, linked, and executed, the following output results:
$x = 1\ y = 2\ z = 3\ w = 4\ v = 5$
Assuming that this program compiled and linked successfully, and based on the output shown above, add static and/or extern modifiers to the blank lines (1) through (10).

Leave them blank if neither modifier would be appropriate.

For example, if we say line (2) is static then we consider line (2) as "static int x". Or if we say line (9) is empty then we treat line 9 as "$\text{int w} = 4$".

A.
Line(1) – empty
Line(2) - static
Line(3) - empty
Line(4) - extern
Line(5) – empty
Line(6) - empty
Line(7) - empty
Line(8) - static
Line(9) - empty
Line(10) - static

B.

Line(1) – static
Line(2) - static
Line(3) - extern
Line(4) - empty
Line(5) – extern
Line(6) - empty
Line(7) - extern
Line(8) - static
Line(9) - empty
Line(10) - static

C.

Line(1) – static
Line(2) - static
Line(3) - empty
Line(4) - empty
Line(5) – empty
Line(6) - empty
Line(7) - empty
Line(8) - static
Line(9) - empty
Line(10) - static

D.

Line(1) – empty
Line(2) - static
Line(3) - empty
Line(4) - empty
Line(5) – extern
Line(6) - empty
Line(7) - empty
Line(8) - static
Line(9) - empty
Line(10) - static

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following variable and function definitions:

```
    int g=10;
    int q3() {
        static int g=5;
        return ++g;
5.  }
    int q4() {
        extern int g;
        return ++g;
    }
10. int q5() {
        int g=1;
        return ++g;
    }
```

What is the value of the expression $q3() + q3() + q4() + q4() + q5() + q5()$?

A. 32
B. 34
C. 38
D. 40

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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